

Task 1 – data modeling

Structure Relationships Calculations Calendars										
Customer_ID	Age	Gender	Region	Income_Level	Membership_Since	Preferred_Channel	member_ship furation (days)	membership duration(years)	CLV	
CUST0001	19	Female	Central	High	30 December 2022	Online	1239	3.39	264.23	
CUST0003	21	Female	South	Medium	10 September 2025	Online	254	0.70	957.67	
CUST0008	63	Female	Central	Medium	05 September 2024	Online	624	1.71	704.47	
CUST0009	42	Female	Central	Low	04 March 2019	Online	2636	7.22	128.93	
CUST0012	37	Female	South	High	27 November 2023	Store	907	2.48	863.59	
CUST0017	41	Female	Central	Medium	30 October 2020	Store	2030	5.56	466.17	
CUST0022	31	Female	West	Low	31 January 2024	Store	842	2.31	1351.71	
CUST0024	31	Female	South	Medium	03 September 2023	Store	992	2.72	1055.37	
CUST0031	68	Female	North	High	03 June 2024	Online	718	1.97	1756.88	
CUST0032	40	Female	South	Low	14 October 2021	Store	1681	4.61	223.51	
CUST0033	55	Female	East	Medium	10 July 2022	Online	1412	3.87	472.45	
CUST0035	21	Female	North	Medium	21 May 2025	Online	366	1.00	581.71	
CUST0039	24	Female	Central	High	26 January 2021	Online	1942	5.32	104.89	
CUST0040	47	Female	South	Medium	17 June 2020	Store	2165	5.93	297.70	
CUST0042	45	Female	West	High	17 May 2023	Online	1101	3.02	710.39	
CUST0043	54	Female	East	Low	25 August 2021	Online	1731	4.74	137.83	
CUST0045	20	Female	East	Medium	27 October 2020	Store	2033	5.57	268.32	
CUST0050	64	Female	East	Low	15 May 2018	Store	2929	8.02	259.01	
CUST0055	41	Female	West	Low	04 September 2017	Store	3182	8.72	108.95	
CUST0059	56	Female	South	High	11 November 2022	Online	1288	3.53	342.17	
CUST0063	44	Female	North	High	25 February 2021	Store	1912	5.24	220.53	
CUST0064	37	Female	North	High	11 September 2022	Store	1349	3.70	316.20	
CUST0065	48	Female	South	High	12 January 2021	Online	1956	5.36	110.09	

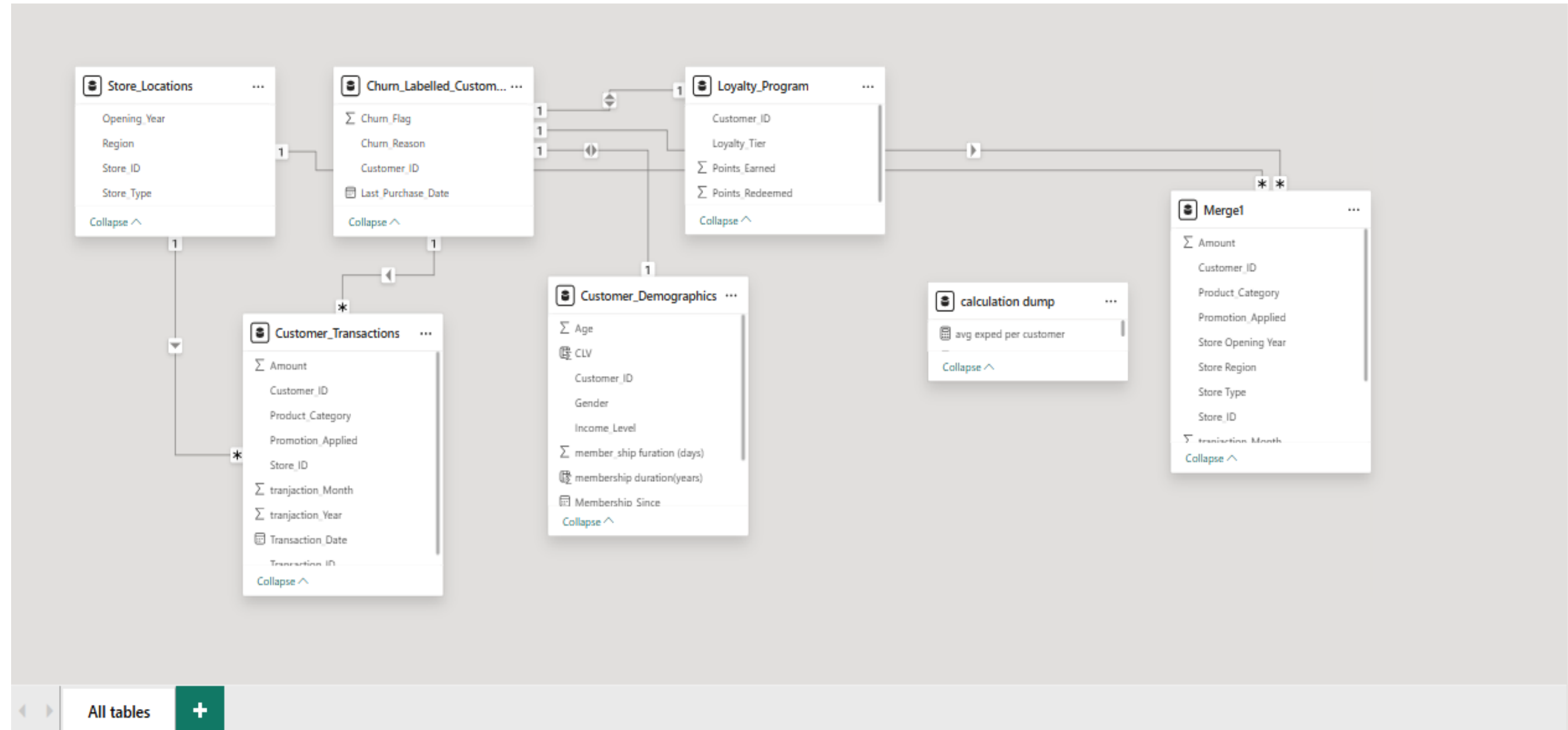
Table: Customer Demographics (300 rows)

Added membership duration column in customers demographics table

Transaction_ID	Customer_ID	Store_ID	Product_Category	Transaction_Date	Amount	Promotion_Applied	tranjaction_Year	tranjaction_Month
TXN00005	CUST0209	STORE041	Groceries	29 March 2024	941.99	Yes	2024	3
TXN00015	CUST0005	STORE037	Groceries	19 April 2024	852.34	No	2024	4
TXN00020	CUST0145	STORE030	Groceries	24 May 2024	689.04	Yes	2024	5
TXN00037	CUST0272	STORE045	Groceries	05 February 2024	137.5	Yes	2024	2
TXN00059	CUST0116	STORE046	Groceries	09 May 2024	591.17	Yes	2024	5
TXN00066	CUST0132	STORE005	Groceries	14 July 2024	759.61	Yes	2024	7
TXN00073	CUST0026	STORE021	Groceries	15 March 2024	408.12	No	2024	3
TXN00074	CUST0295	STORE007	Groceries	19 June 2024	289.74	Yes	2024	6
TXN00095	CUST0196	STORE026	Groceries	10 September 2024	692	No	2024	9
TXN00099	CUST0041	STORE045	Groceries	18 March 2024	662.25	Yes	2024	3
TXN00118	CUST0272	STORE005	Groceries	25 November 2024	375.09	Yes	2024	11
TXN00119	CUST0027	STORE007	Groceries	10 April 2024	775.26	Yes	2024	4
TXN00126	CUST0099	STORE025	Groceries	16 December 2024	743.12	Yes	2024	12
TXN00128	CUST0015	STORE017	Groceries	09 April 2024	631.15	No	2024	4
TXN00150	CUST0128	STORE037	Groceries	27 May 2024	337.59	No	2024	5
TXN00151	CUST0042	STORE045	Groceries	25 February 2024	539.38	No	2024	2
TXN00165	CUST0012	STORE035	Groceries	30 January 2024	886.53	Yes	2024	1
TXN00166	CUST0157	STORE022	Groceries	22 May 2024	102.04	No	2024	5
TXN00168	CUST0250	STORE024	Groceries	05 February 2024	169.27	No	2024	2
TXN00169	CUST0224	STORE017	Groceries	11 September 2024	662.04	No	2024	9
TXN00170	CUST0112	STORE006	Groceries	29 December 2024	775.49	No	2024	12
TXN00191	CUST0225	STORE006	Groceries	24 March 2024	18.52	No	2024	3
TXN00193	CUST0177	STORE010	Groceries	09 Februarv 2024	756.79	No	2024	2

Extrcted transaction month and year in transaction table

Final model view of the dataset



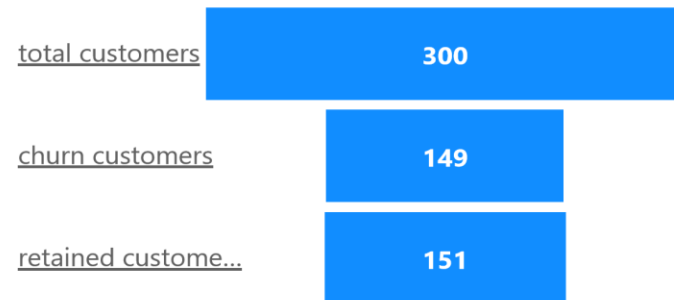
Task 2 – Churn and retention metrics



Walmart

Funnel chart

retained and churned customers



300

total customers

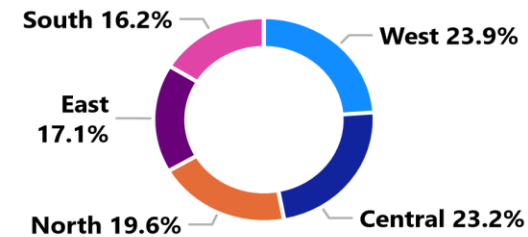
149

churn customers

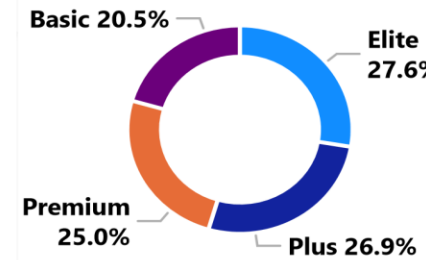
49.67%

churn rate

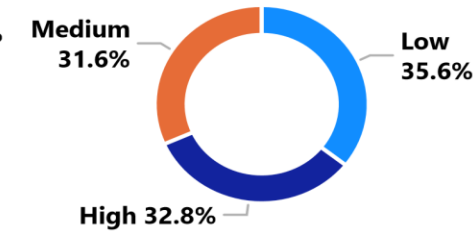
**churn rate
by Region**



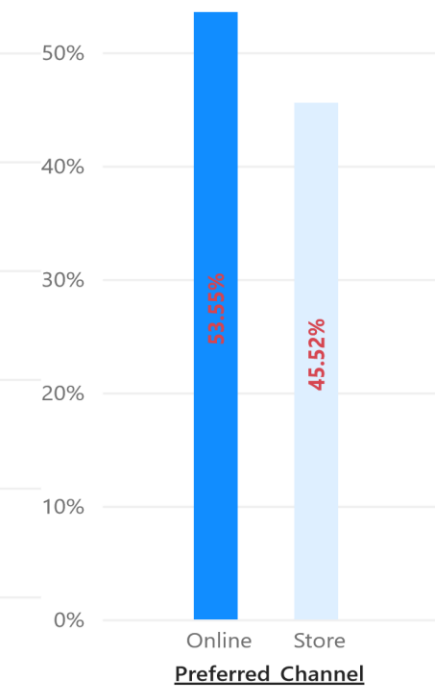
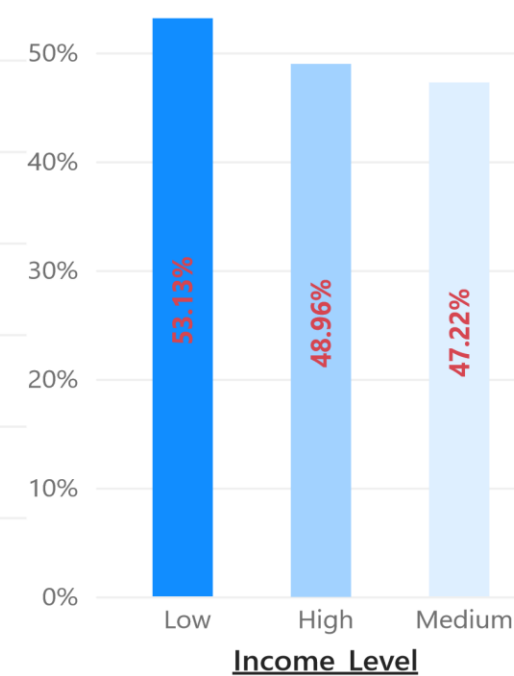
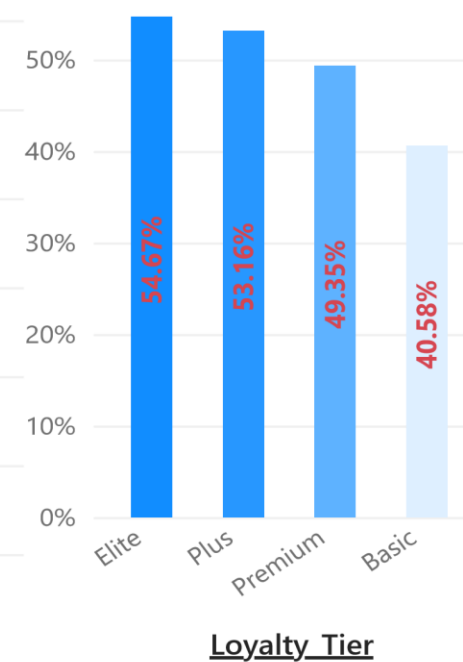
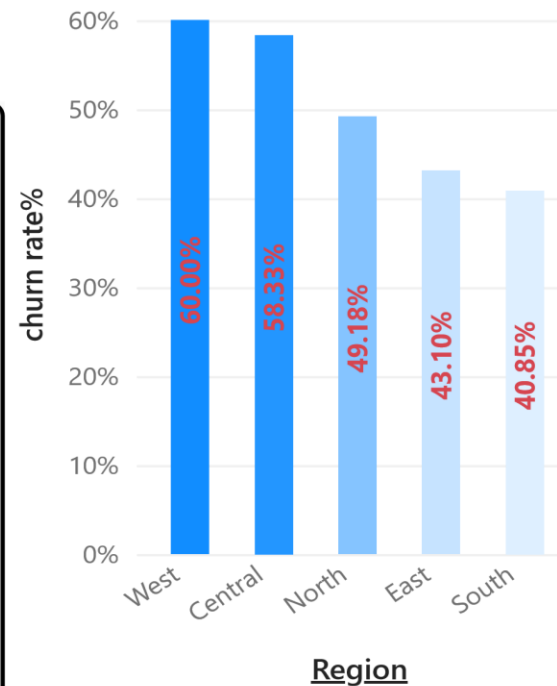
**churn rate
by Loyalty_Tier**



**churn rate
by Income_Level**



**churn rate
by Preferred_Channel**



Task 3 - Repeat purchase analysis



low tier customers

Customer_ID	tranjection count
CUST0006	3
CUST0007	3
CUST0012	3
CUST0014	3
CUST0021	3
CUST0023	3
CUST0025	3
CUST0027	3
CUST0033	3
CUST0035	3
CUST0039	3
CUST0046	3

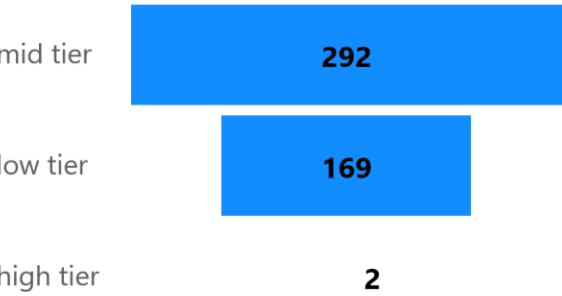
mid tier customers

Customer_ID	tranjection count
CUST0219	8
CUST0041	7
CUST0049	7
CUST0057	7
CUST0079	7
CUST0114	7
CUST0120	7
CUST0127	7
CUST0135	7
CUST0137	7
CUST0142	7
CUST0185	7

High tier customers

Customer_ID	Count of Transaction_ID
CUST0136	9
CUST0200	9

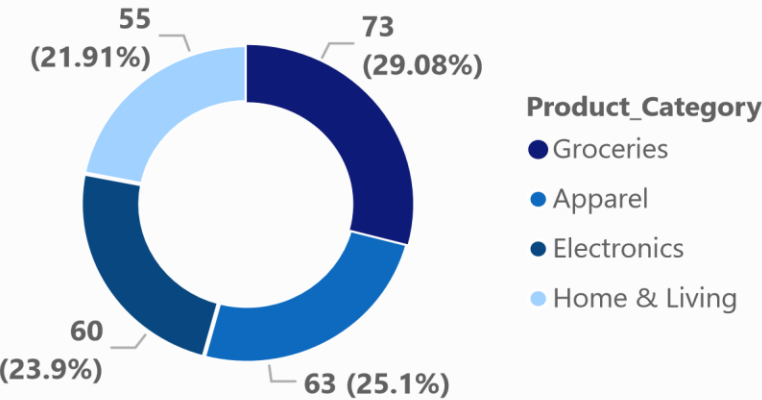
No. of customers in



most purchased product category
by premium customers

Groceries

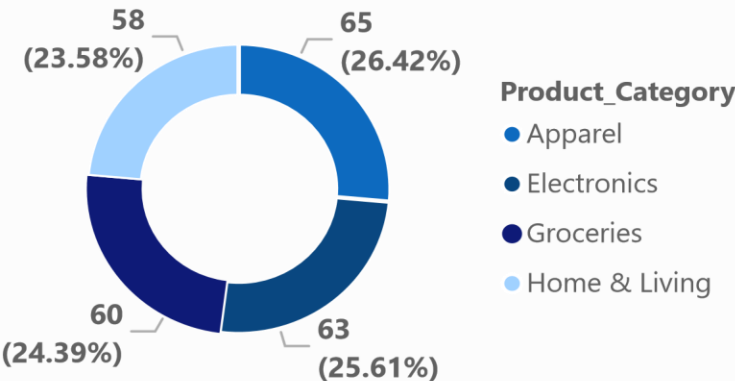
No. of Transactions(premium customers)
by Product_Category



most purchased product category
by elite customers

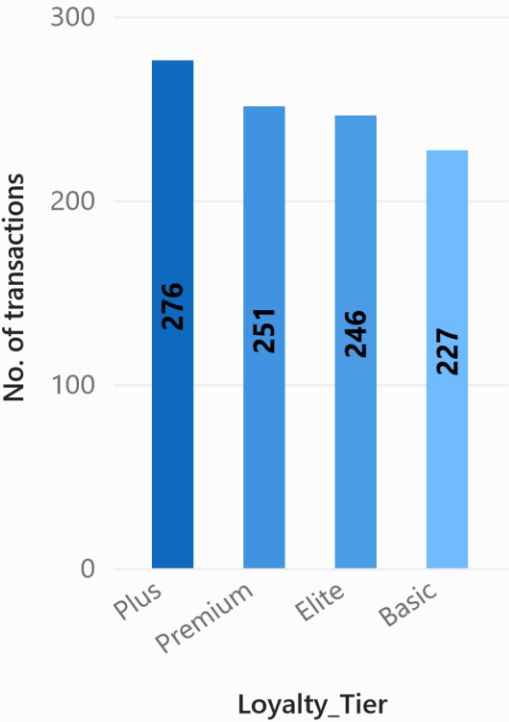
Apparel

no. of Transaction(Elite customers)
By Product Category



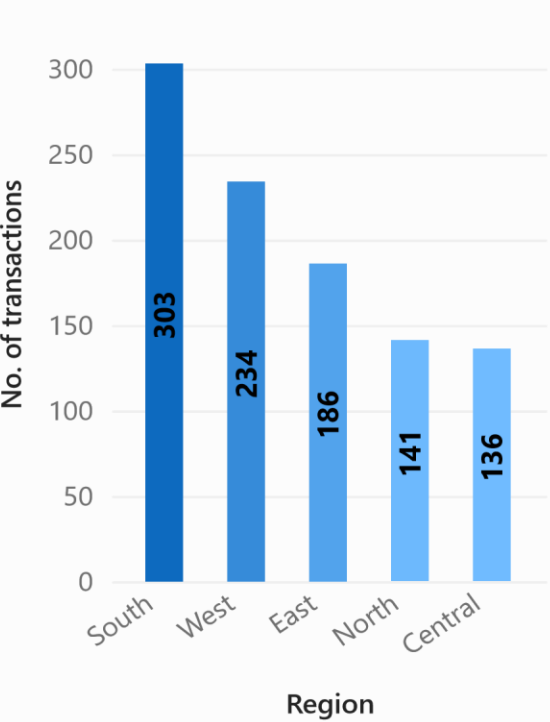
frequency of Transactions

By Loyalty_Tier



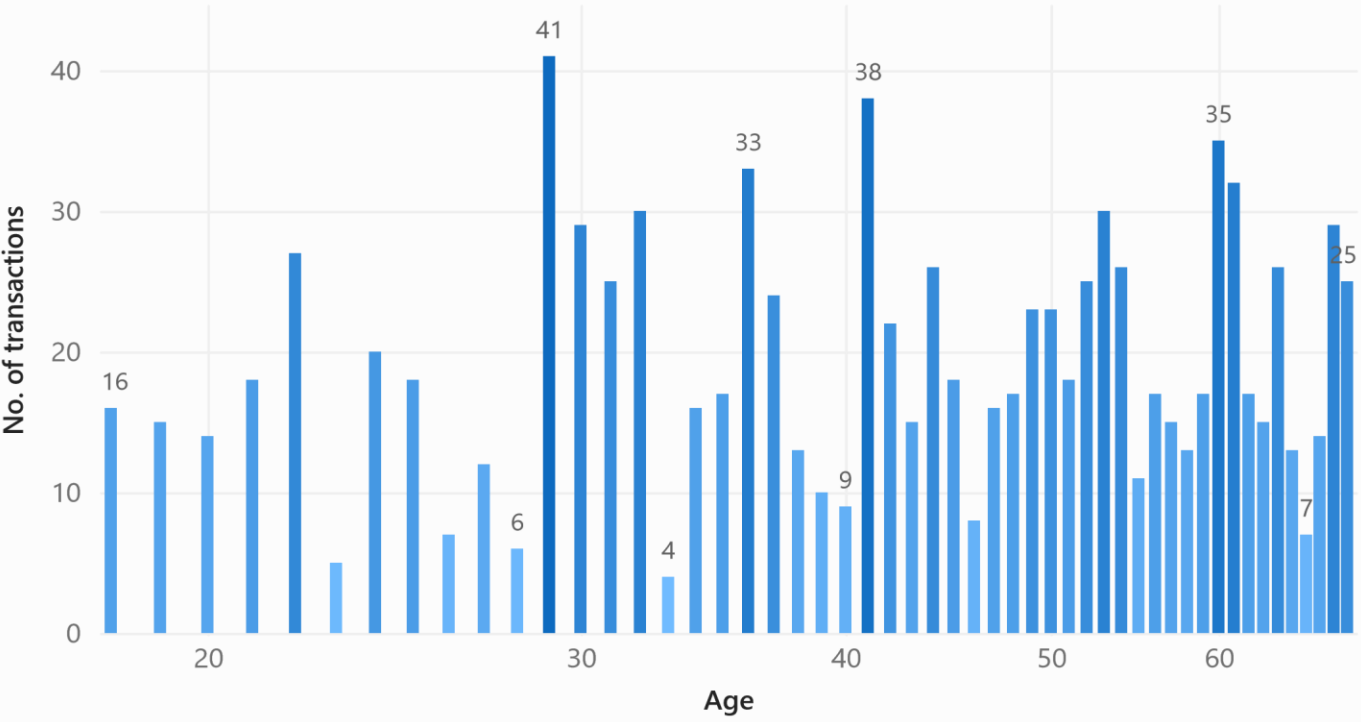
frequency of Transactions

By Region

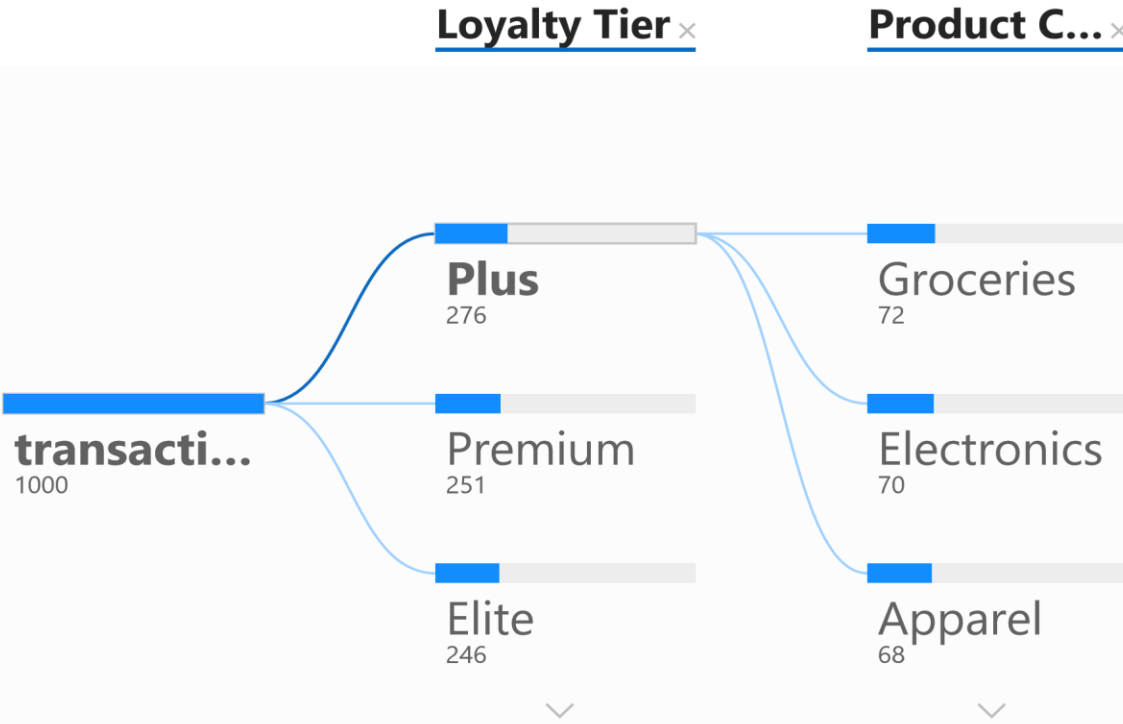


frequency of Transaction

By Age



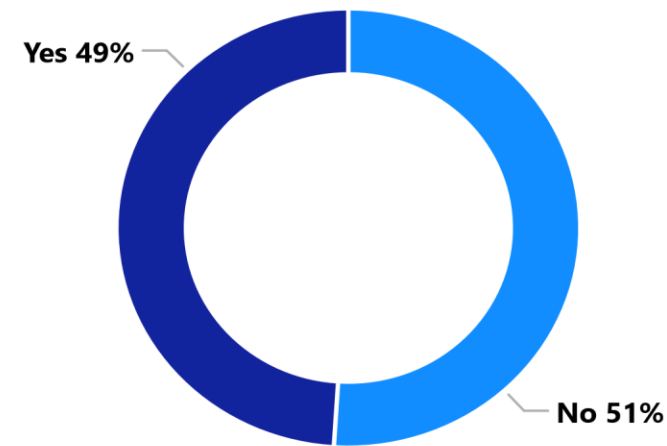
product categories purchased by customers grouped by loyalty.



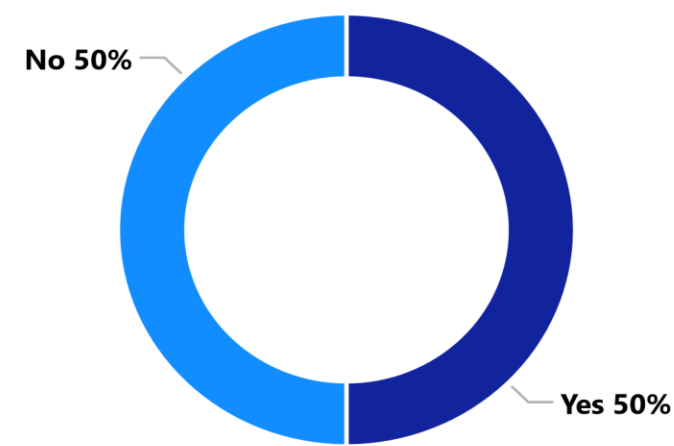
Task 4 - Promotion and Loyalty impact



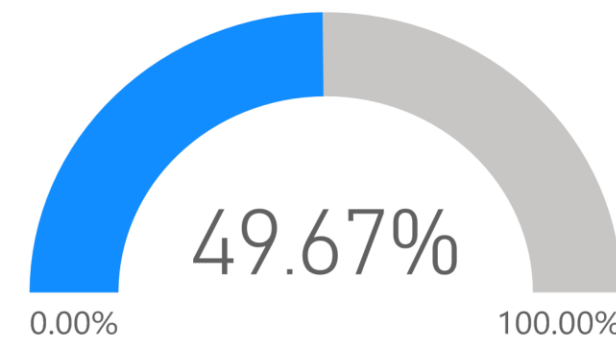
No. of Transactions
by Promotion_Applied



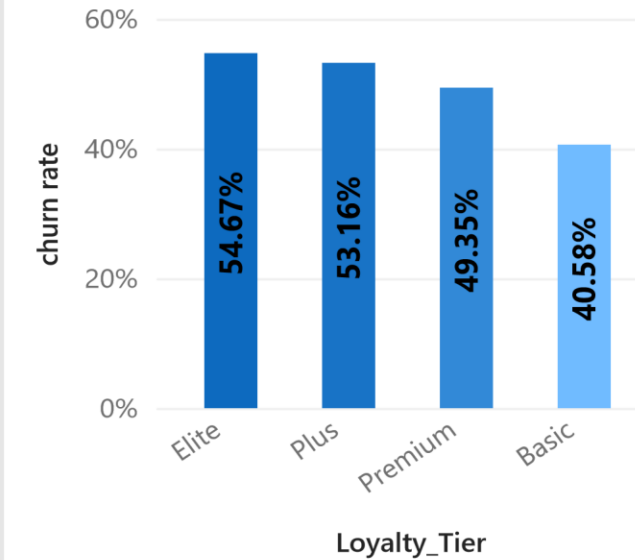
Average of Amount
by Promotion_Applied



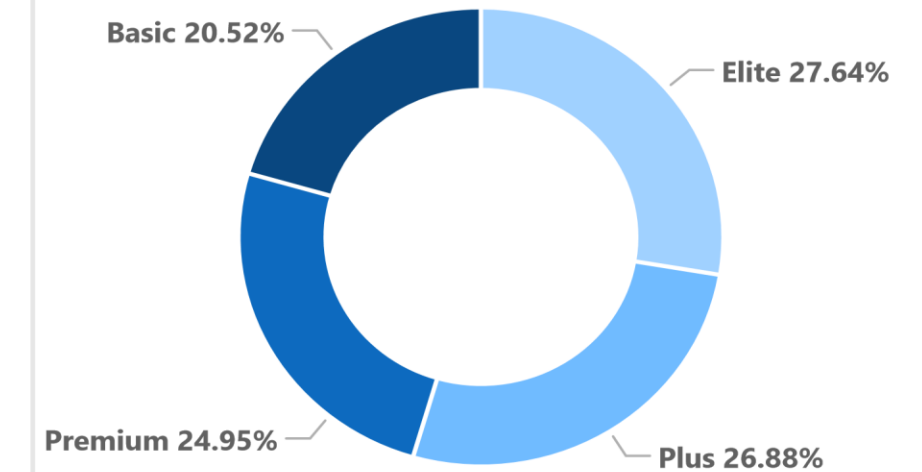
churn rate



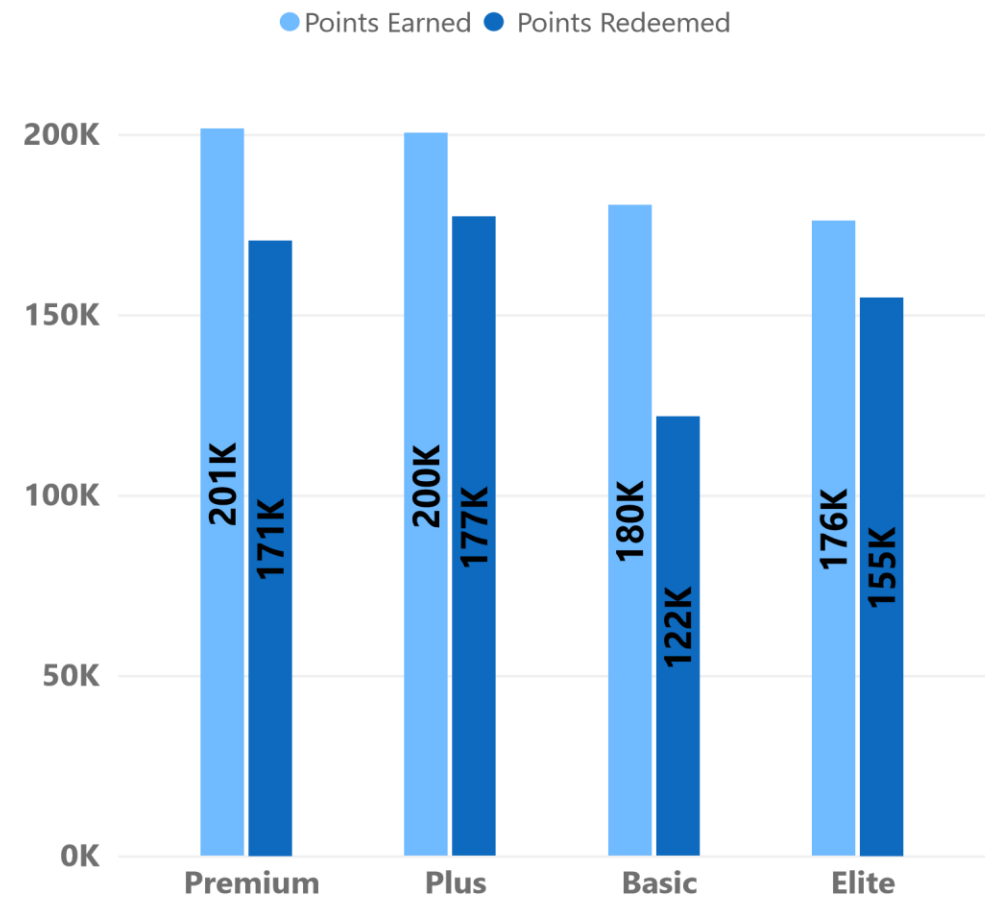
churn rate
by Loyalty_Tier



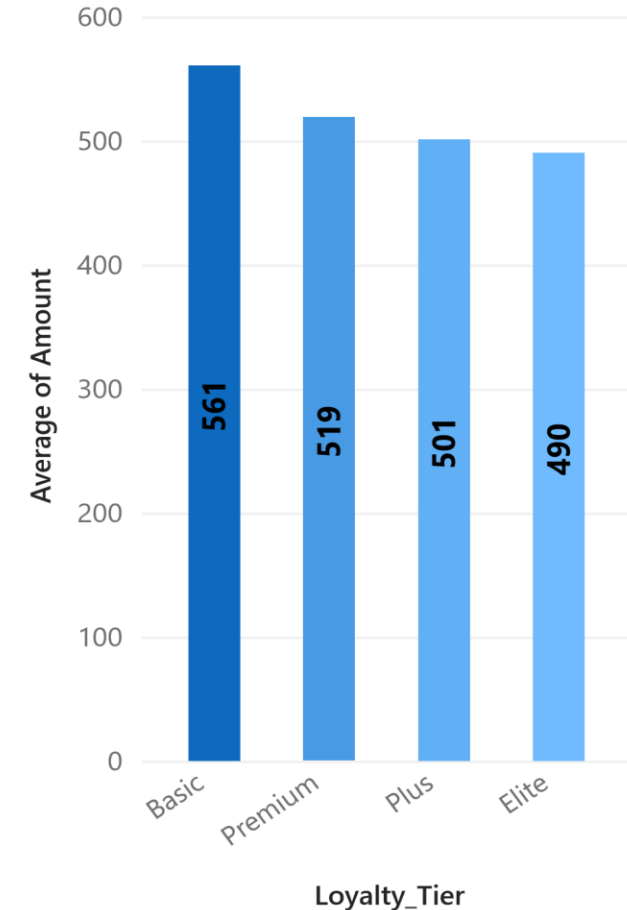
churn rate
by Loyalty Tier (% of total)



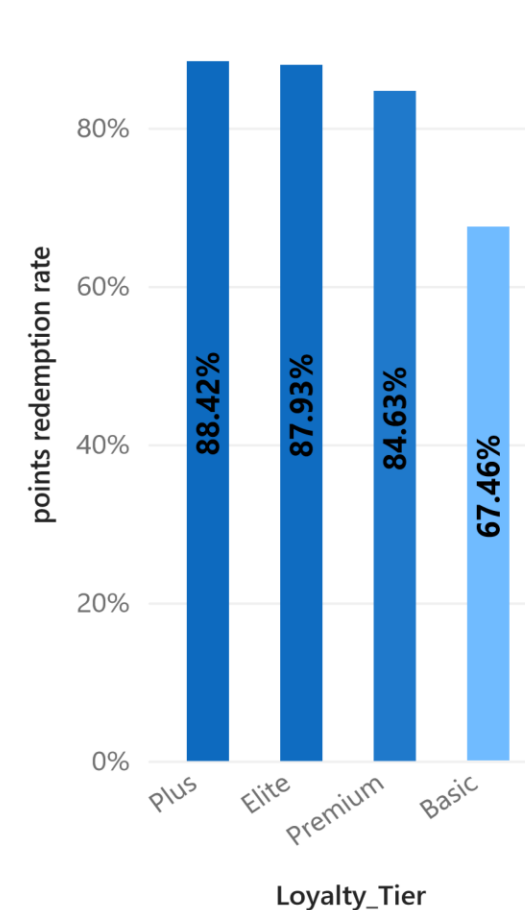
Points Earned and Points Redeemed
by Loyalty_Tier



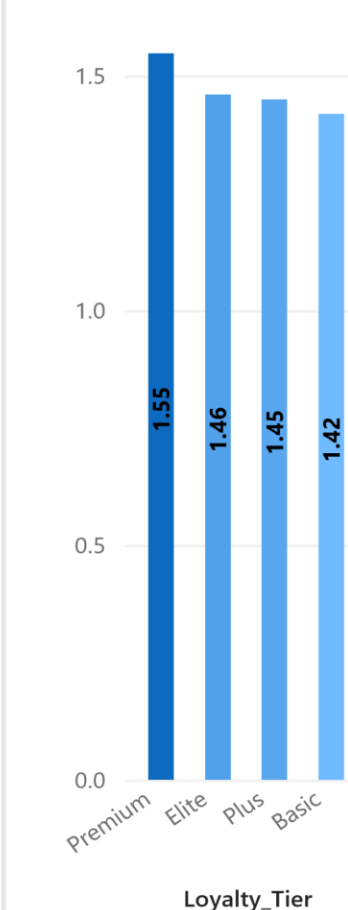
Average of Amount spent
by Loyalty_Tier



points redemption rate
by Loyalty_Tier



point earn rate
by Loyalty_Tier



RECOMMENDATION TO IMPROVE REDEMPTION AND RETENTION

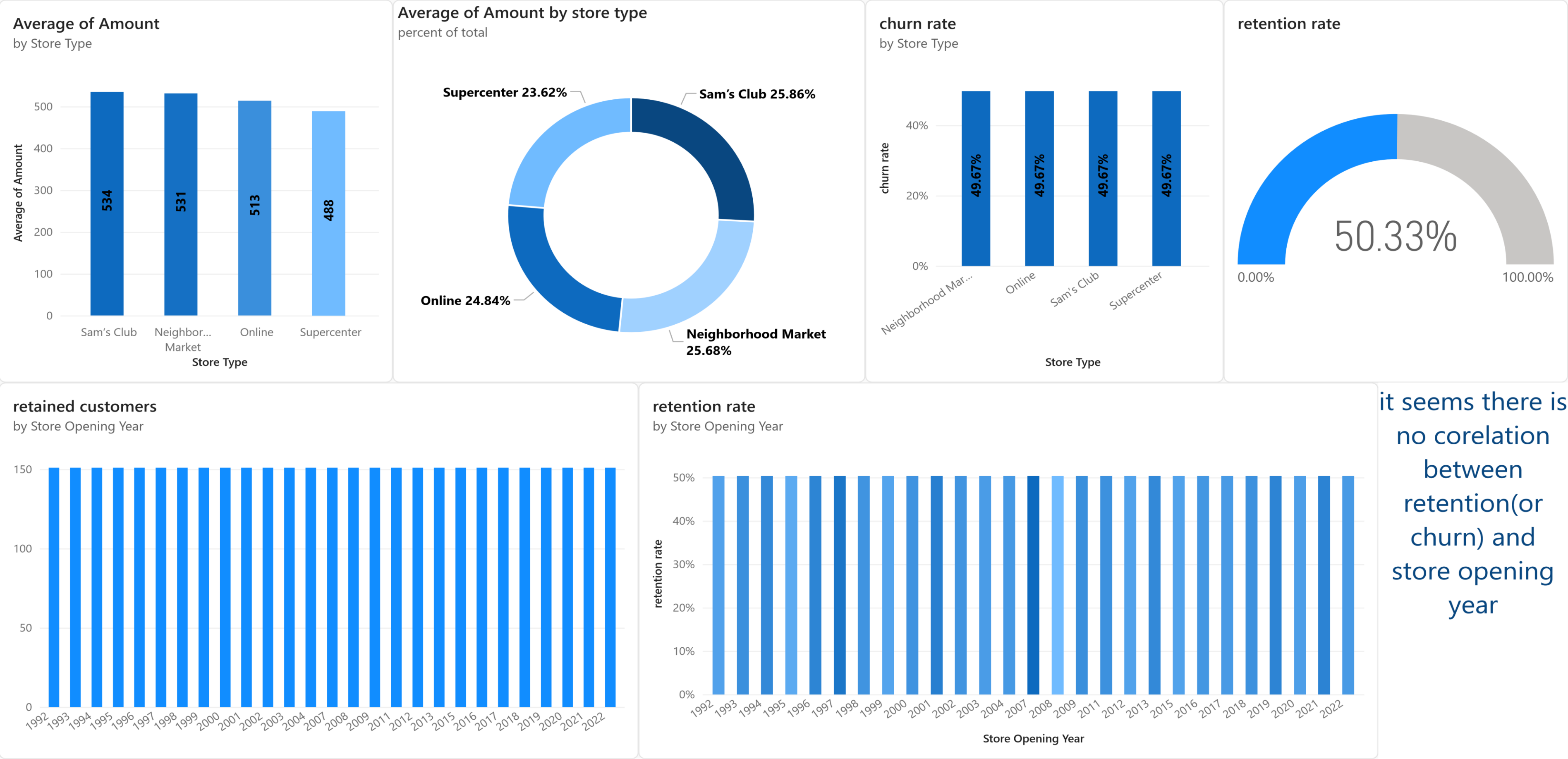
As we can see that churn rate is lowest and points redeemed(out of total points earned) is lowest also points earned (on each rupee spent) is lowest among basic tier customers. Also basic tier customers spend more (on an avg.)

step 1. distribute equal amount of points based on expenditure among all customers

step 2. promote using points among basic customers

this way more they will spend more they will earn points , then they will again redeem the points and purchase more and more , in this loop they will redeem more points and become more loyal hence redemption and retention both will improve

Task 5 – Store and channel performance vs Retention



NameMerge1

Manage relationships

Relationships

New measure

Quick measure

New column

New table

Calculations

Mark as date table

Calendars

Structure

×

✓

Structure

Table

Column

Relationship

Measure

Table

Column

Relationship

Measure

Transaction_ID	Customer_ID	Store_ID	Product_Category	Transaction_Date	Amount	Promotion_Applied	tranjaction_Year	tranjaction_Month	Store Type	Store Region	Store
TXN00015	CUST0005	STORE037	Groceries	19 April 2024	852.34	No	2024	4	Online	South	20
TXN00016	CUST0159	STORE023	Electronics	23 June 2024	314.76	Yes	2024	6	Online	South	19
TXN00021	CUST0046	STORE046	Home & Living	05 July 2024	995.81	Yes	2024	7	Online	South	20
TXN00043	CUST0049	STORE046	Apparel	16 March 2024	947.15	Yes	2024	3	Online	South	20
TXN00038	CUST0075	STORE043	Electronics	13 February 2024	511.48	Yes	2024	2	Online	South	20
TXN00057	CUST0169	STORE023	Apparel	04 October 2024	392.47	Yes	2024	10	Online	South	19
TXN00059	CUST0116	STORE046	Groceries	09 May 2024	591.17	Yes	2024	5	Online	South	20
TXN00067	CUST0008	STORE031	Apparel	03 August 2024	749.25	Yes	2024	8	Online	South	20
TXN00096	CUST0130	STORE023	Home & Living	09 September 2024	205.03	No	2024	9	Online	South	19
TXN00107	CUST0107	STORE046	Electronics	30 June 2024	143.98	Yes	2024	6	Online	South	20
TXN00114	CUST0028	STORE010	Electronics	14 January 2024	814.16	Yes	2024	1	Online	South	20
TXN00121	CUST0201	STORE037	Electronics	08 January 2024	502.56	No	2024	1	Online	South	20
TXN00150	CUST0128	STORE037	Groceries	27 May 2024	337.59	No	2024	5	Online	South	20
TXN00184	CUST0071	STORE031	Apparel	27 October 2024	264.61	Yes	2024	10	Online	South	20
TXN00185	CUST0230	STORE046	Apparel	12 August 2024	726.69	Yes	2024	8	Online	South	20
TXN00193	CUST0177	STORE010	Groceries	09 February 2024	756.79	No	2024	2	Online	South	20
TXN00202	CUST0274	STORE046	Groceries	03 February 2024	156.76	Yes	2024	2	Online	South	20
TXN00209	CUST0154	STORE043	Groceries	07 March 2024	420.68	No	2024	3	Online	South	20
TXN00239	CUST0024	STORE043	Home & Living	26 December 2024	279.83	Yes	2024	12	Online	South	20
TXN00258	CUST0073	STORE023	Home & Living	08 December 2024	412.34	No	2024	12	Online	South	19
TXN00264	CUST0038	STORE031	Electronics	12 February 2024	939.77	Yes	2024	2	Online	South	20
TXN00280	CUST0288	STORE046	Groceries	18 June 2024	363.33	No	2024	6	Online	South	20

Table

Column

Relationship

Measure

Table

Column

Relationship

Measure

Expand

Store and transaction dataset merged

Task 6 – Customers lifetime value (CLV) analysis



NO. of customers
BY CLV



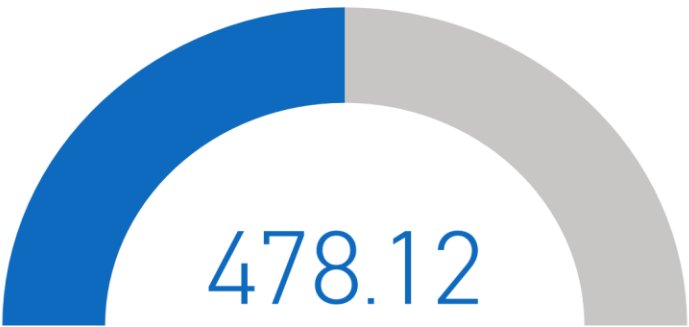
High value customers

Customer_ID	CLV
CUST0037	497.26
CUST0026	497.85
CUST0152	504.08
CUST0200	505.39
CUST0066	510.49
CUST0236	513.07
CUST0213	514.33
CUST0114	521.34
CUST0142	529.07
CUST0215	534.79
CUST0053	543.84
CUST0118	548.63

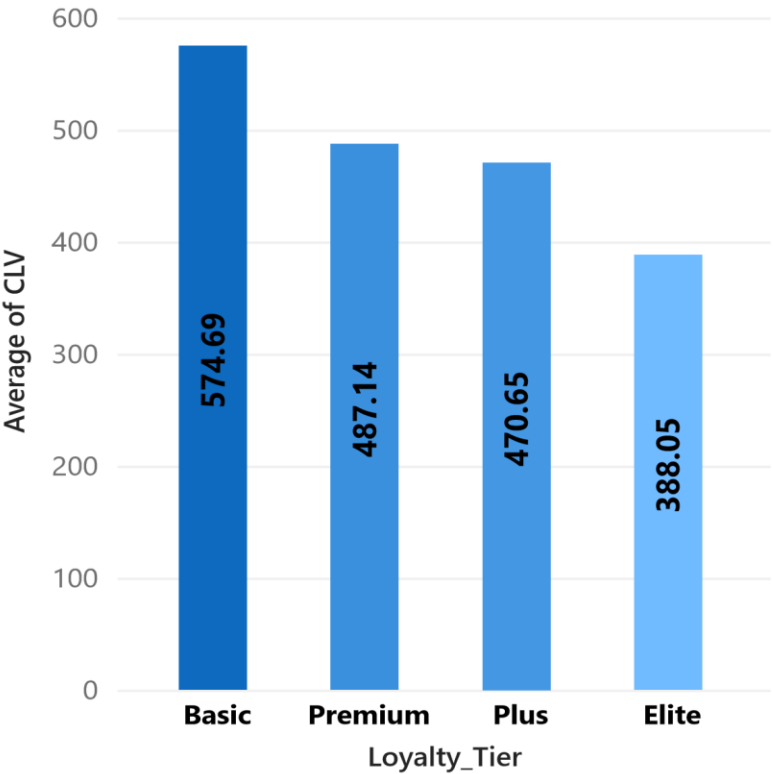
low value customers

Customer_ID	CLV
CUST0300	32.28
CUST0299	342.12
CUST0297	196.17
CUST0296	183.19
CUST0295	216.35
CUST0294	384.80
CUST0293	316.65
CUST0292	281.33
CUST0291	152.07
CUST0290	142.61
CUST0289	117.29
CUST0288	426.26

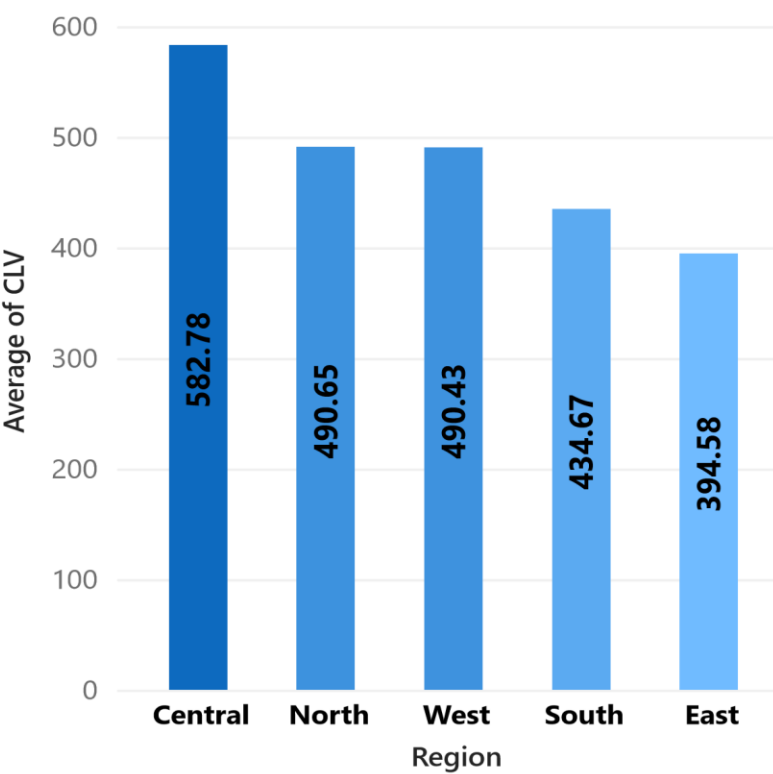
Average CLV
of all the customers



Average of CLV
by Loyalty_Tier



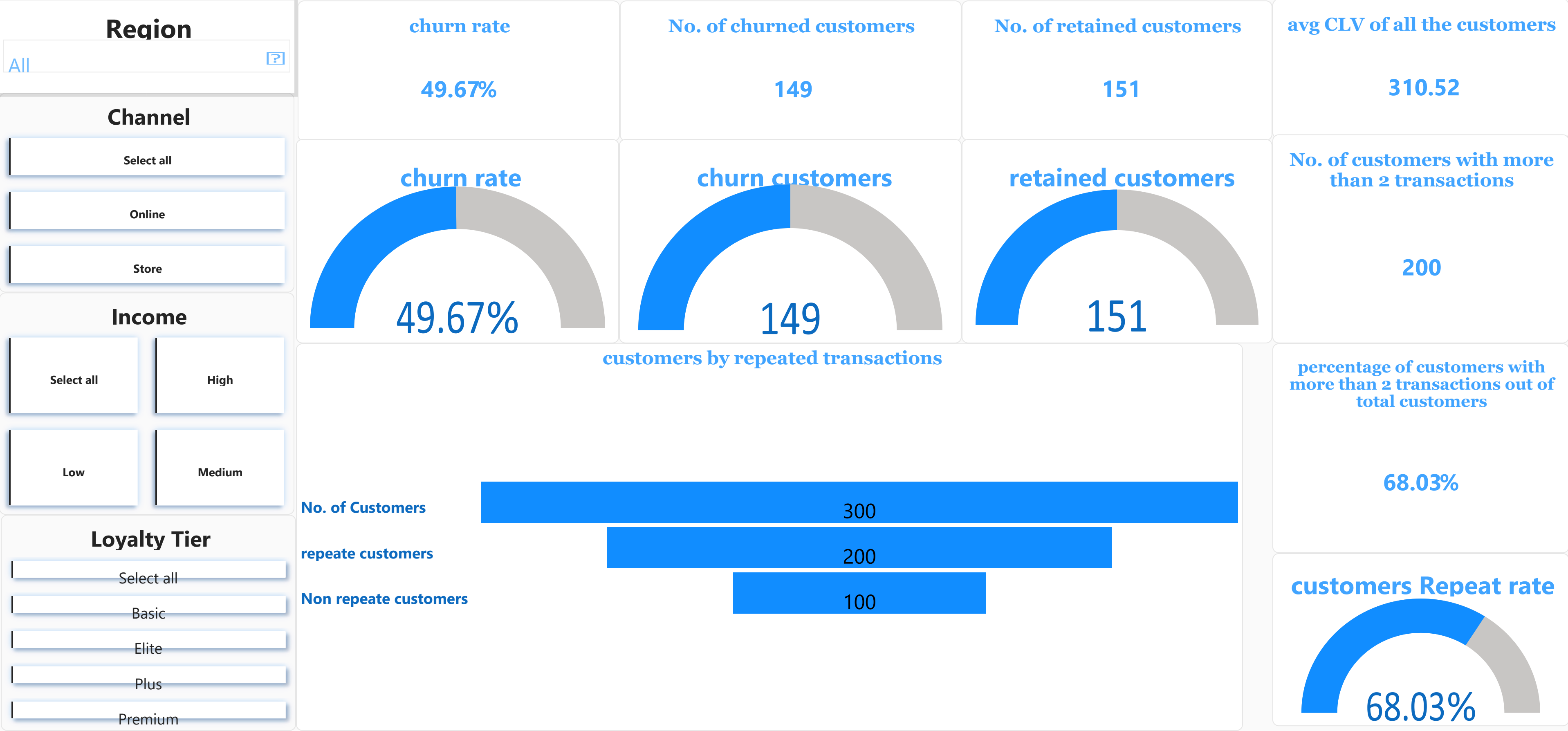
Average of CLV
by Region



Average of CLV
by Last_Purchase_Date

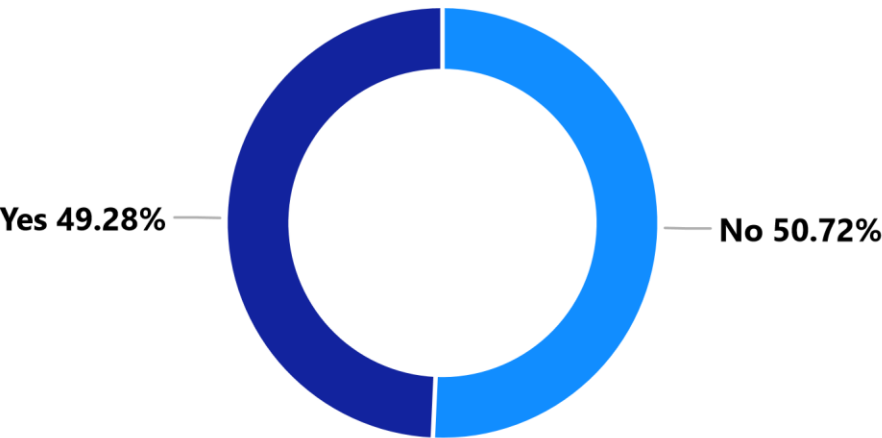


Slicers



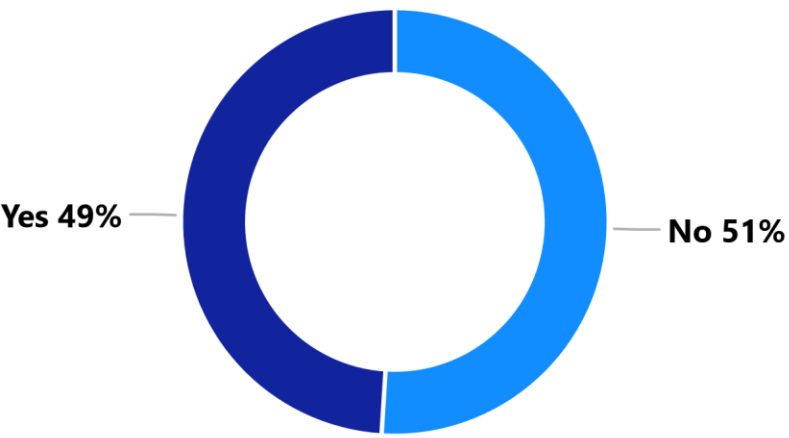
proportion of customers

by promotion applied



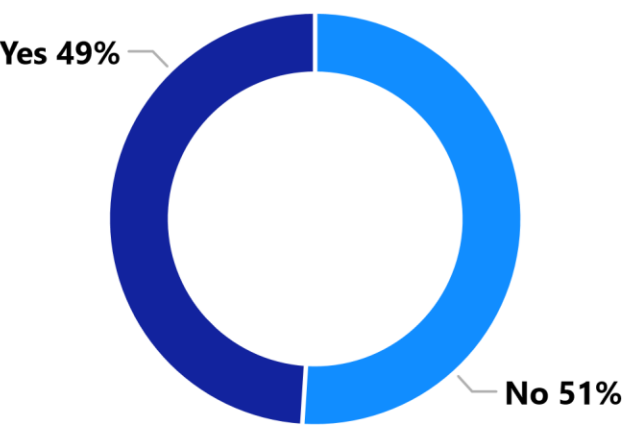
Total amount spent

with and without promotion



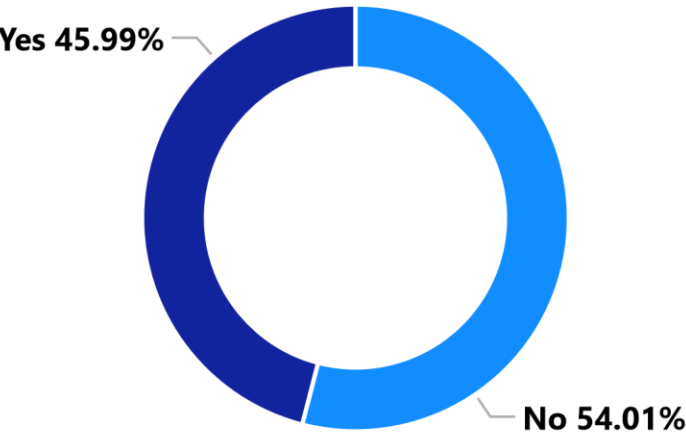
proportion of transaction

with and without promotion applies



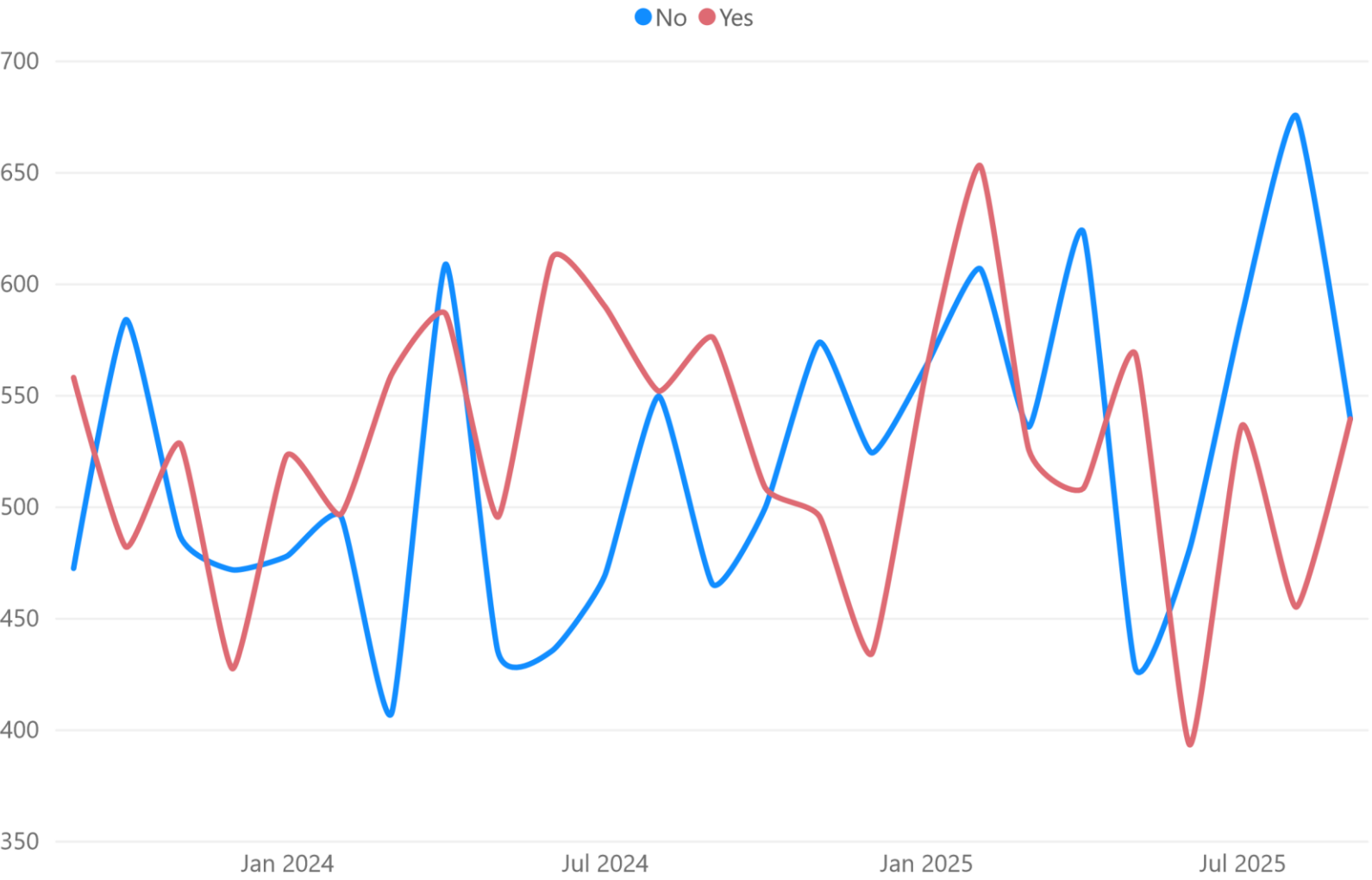
proportion of repeat customers

with and without promotion applied



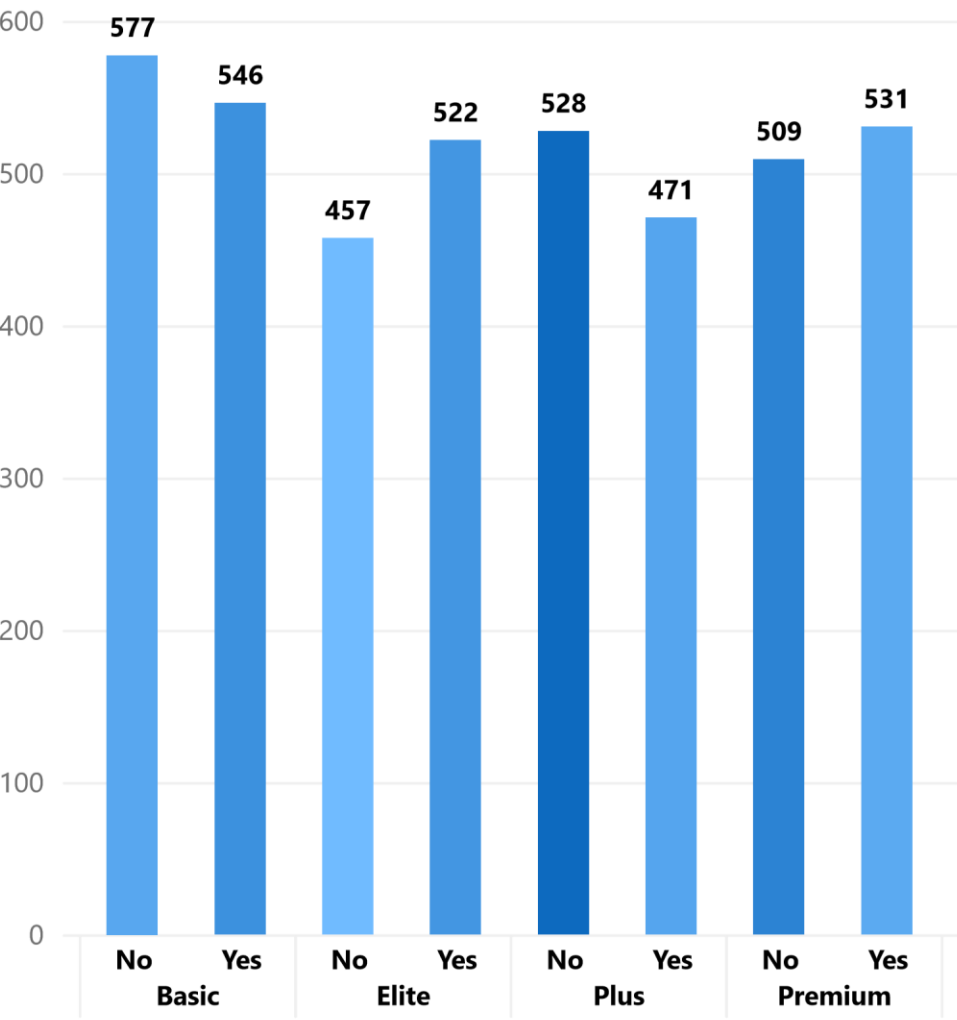
Average of Amount

by Year, Month and Promotion_Applied



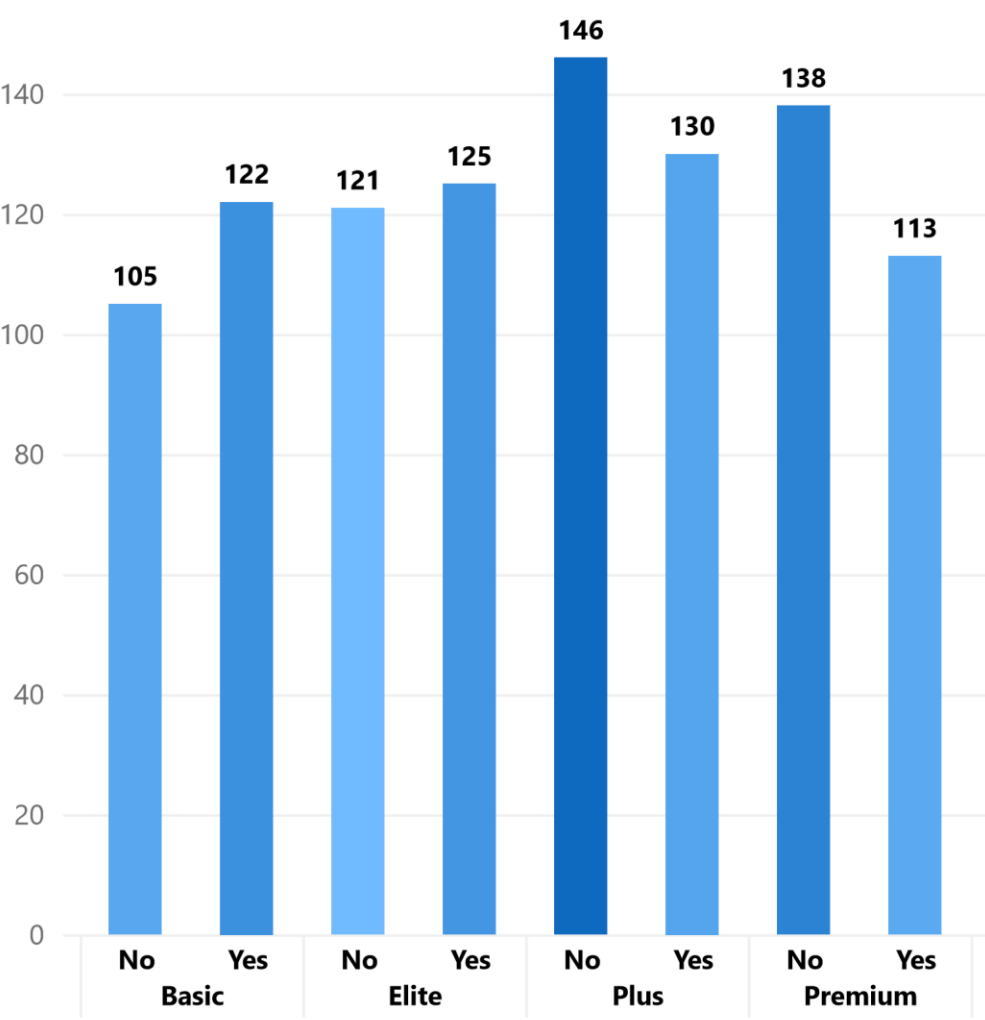
Average Amount spent

by loyalty tier and promotion applied



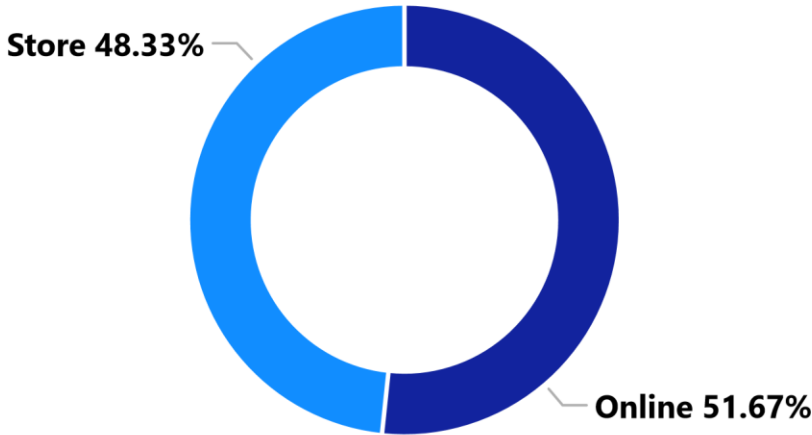
No. of Transactions

by Loyalty_Tier and Promotion_Applied

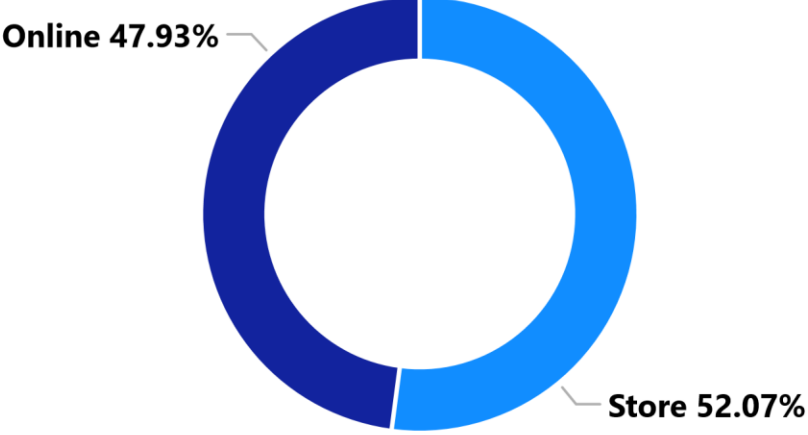


Proportion of Customers

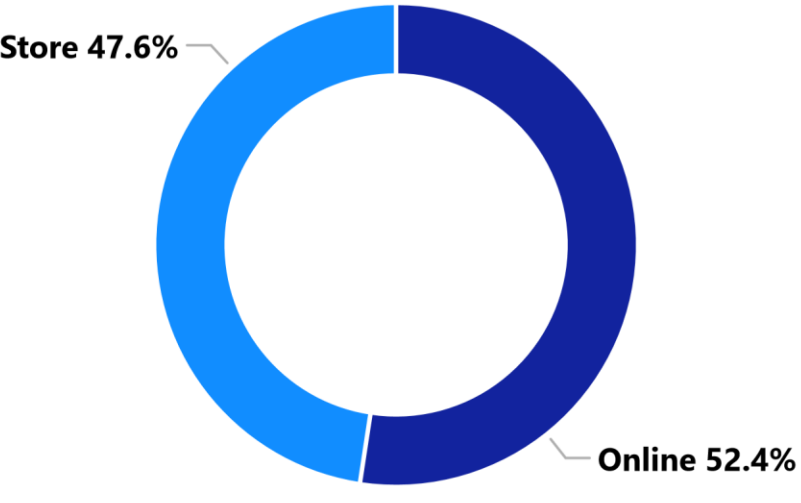
By their Preferred Channel



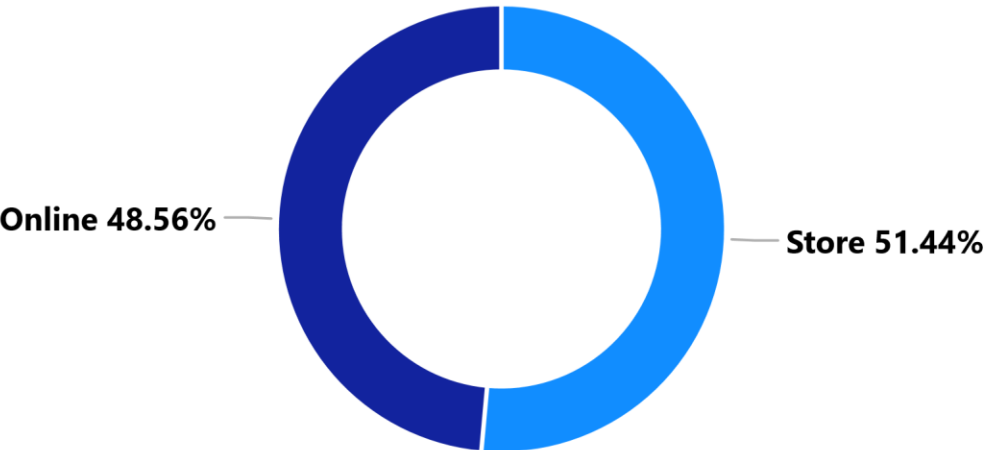
Proportion of Average Amount spent



No. of Transaction

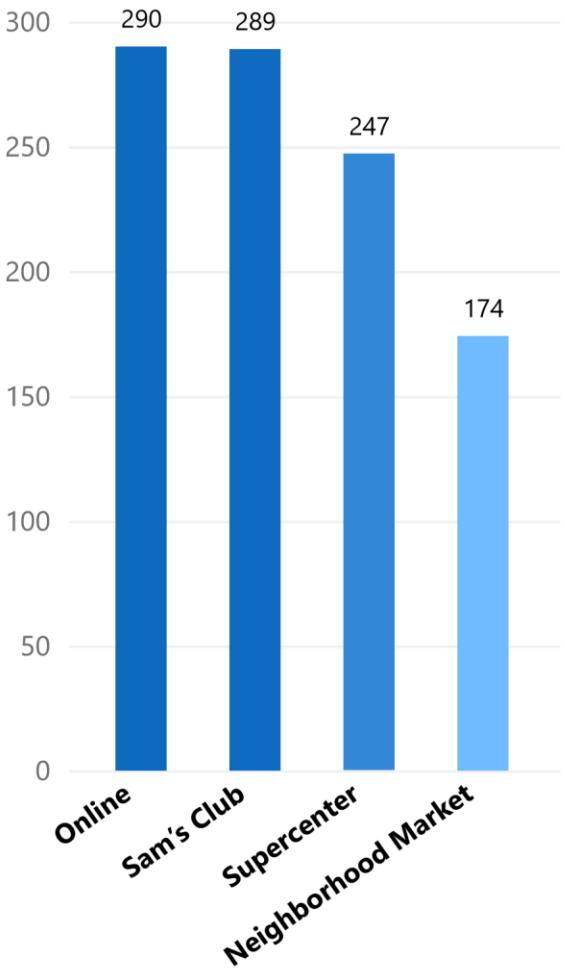


Sum of Points_Earned by Preferred_Channel



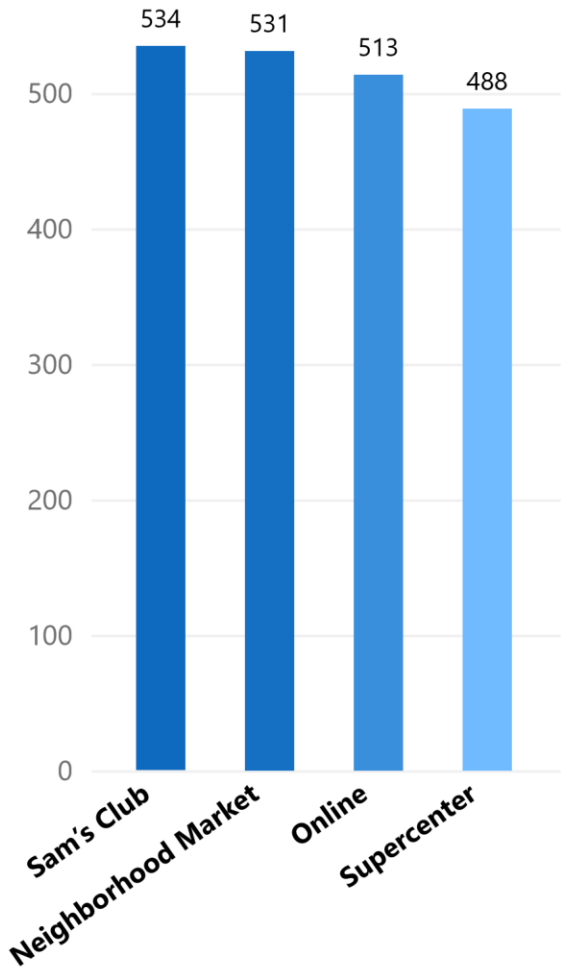
No. of Customers

By Store Type



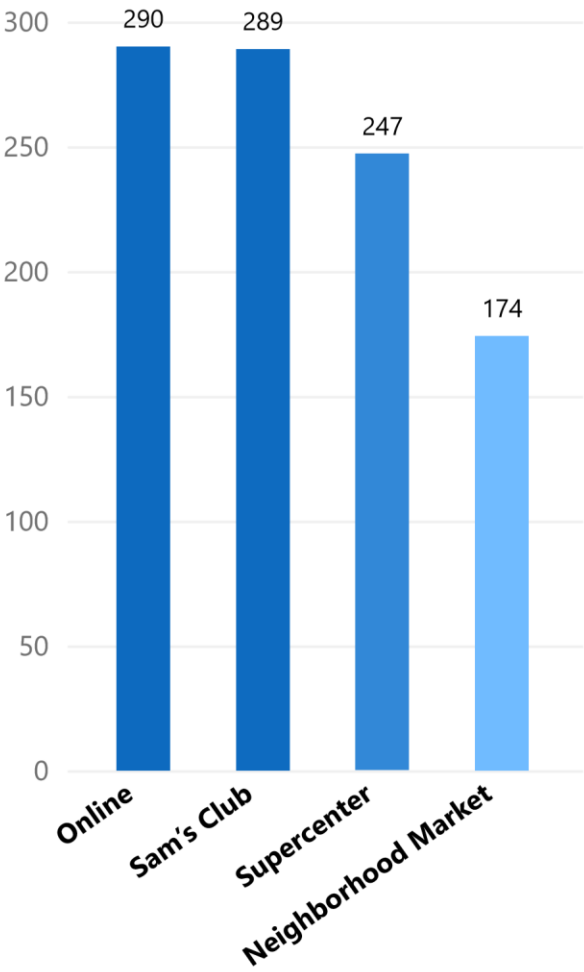
Average of Amount

by Store_Type



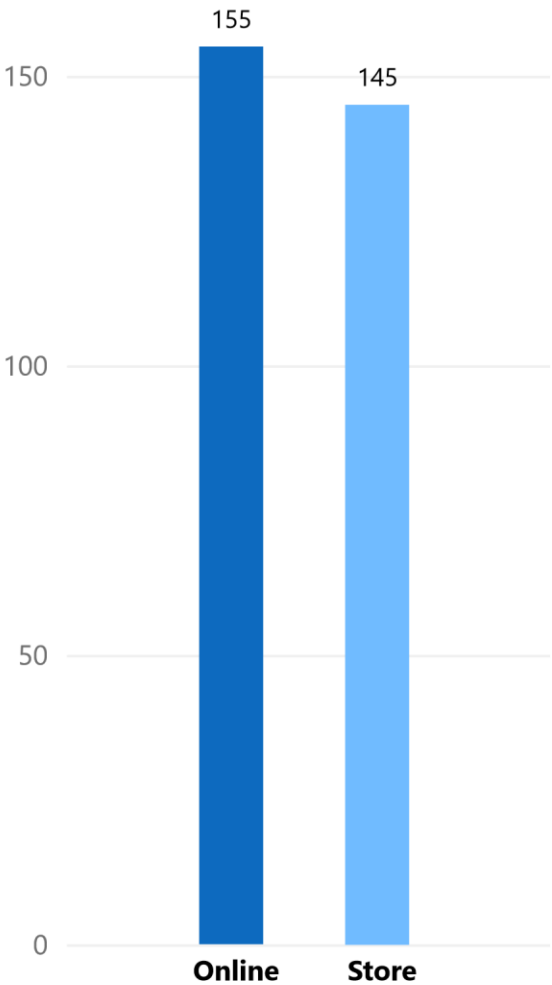
No. of Transactions

by Store_Type



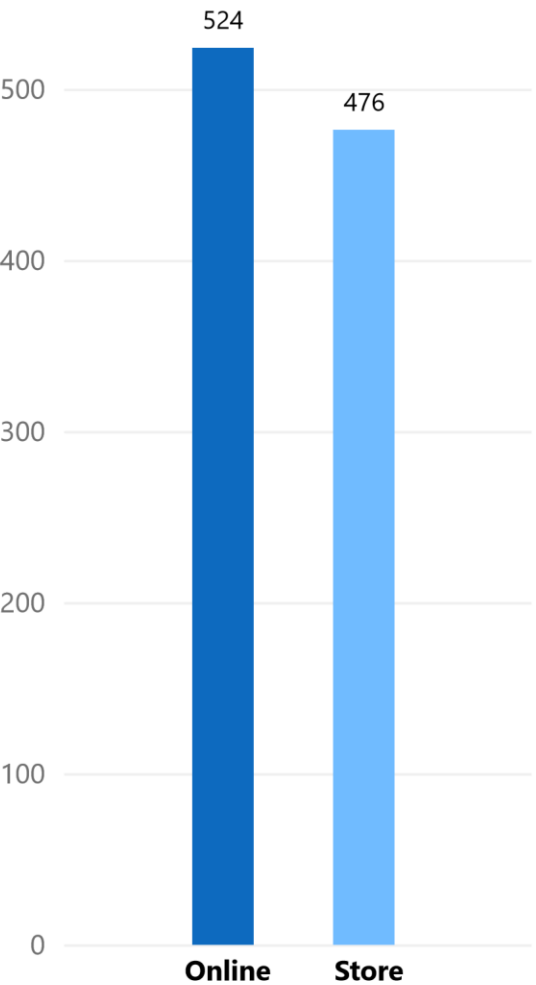
No. of Customers

By Preferred Channel



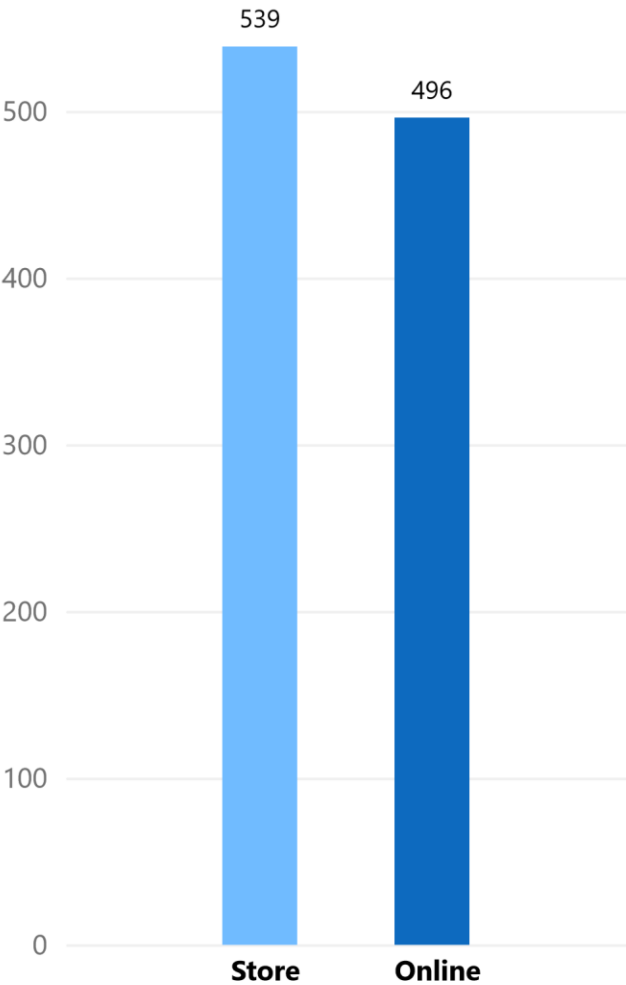
No. of Transactions

by Preferred Channel

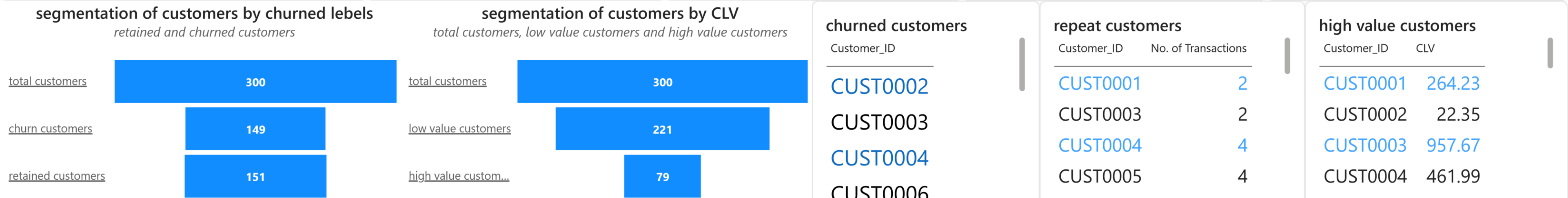


Average Amount spent

by Preferred Channel



No. of total customers	No. of churned customers	No. of customers with repeat transactions	No. of high value customers	No. of low value customers
300	149	200	79	221



churned customers

Customer_ID
CUST0002
CUST0003
CUST0004
CUST0006
CUST0008
CUST0010
CUST0011
CUST0013
CUST0014
CUST0015
CUST0016
CUST0020
CUST0023
CUST0025
CUST0026

repeat customers

Customer_ID	No. of Transactions
CUST0001	2
CUST0003	2
CUST0004	4
CUST0005	4
CUST0006	3
CUST0007	3
CUST0008	2
CUST0010	4
CUST0011	2
CUST0012	3
CUST0013	5
CUST0014	3
CUST0015	4
CUST0016	2
CUST0017	5
CUST0019	5
CUST0020	2

high value customers

Customer_ID	CLV
CUST0001	264.23
CUST0002	22.35
CUST0003	957.67
CUST0004	461.99
CUST0005	383.53
CUST0006	268.90
CUST0007	106.24
CUST0008	704.47
CUST0009	128.93
CUST0010	443.12
CUST0011	829.98
CUST0012	863.59
CUST0013	120.77
CUST0014	225.30
CUST0015	261.03
CUST0016	481.34
CUST0017	466.17

Recommendations

- 1: As seen in task 4 and 6 that on an avg basic customers have highest CLV and avg spending, meaning basic customers spend highest in their total membership duration. so if prioritize basic customers then the revenue of the company will increase so we should prioritize basic customers for retention
- * 2:As per review of page 3 of task 7, highest no. of customers are available on online channel , also the no. of transaction is highest on online channel , but avg amount spent is not as much as no. of customers and transaction , rather on an avg people spend more in online channel , hence online channel is underperforming in generating revenue , while offline channels are underperforming in terms of customers engagement and purchases
- * 3: Basic customers are ranked lowest in terms on loyalty , but as we have seen that their avg spending is more than customers in every other tier , so we can distribute reward points based on spending , and then we should promote the uses of rewards points , this way they will spend more and more and come back again and again , hence it will strengthen loyalty program.

All the measures used in the visuals

- 1.avg expenditure per customer = `DIVIDE(SUM(Customer_Transactions[Amount]),COUNT(Customer_Transactions[Customer_ID]),0)`
- 2.avg_clv = `AVERAGE(Customer_Demographics[CLV])`
- 3.CLV = `DIVIDE(SUMX(RELATEDTABLE(Customer_Transactions), Customer_Transactions[Amount]),Customer_Demographics[membership duration(years)],0)`, made CLV column in Customer Demographics table
4. avg_transaction = `AVERAGEX(Customer_Transactions,COUNT(Customer_Transactions[Transaction_ID]))`
5. churn customers = `CALCULATE(count(Churn_Labelled_Customers[Customer_ID]),Churn_Labelled_Customers[Churn_Flag] =1)`
6. churn_rate=`DIVIDE(CALCULATE(COUNT(Churn_Labelled_Customers[Customer_ID]),Churn_Labelled_Customers[Churn_Flag]=1),COUNT(Churn_Labelled_Customers[Customer_ID]),1)`
7. CLV = `DIVIDE(SUM(Customer_Transactions[Amount]),SUM(Customer_Demographics[membership duration(years)]),0)`
8. customers repeate rate = `DIVIDE([repeate customers],DISTINCTCOUNT(Customer_Transactions[Customer_ID]),0)`
9. most pruchased product by premium customers = `CALCULATE(COUNT(Customer_Transactions[Transaction_ID]),Loyalty_Program[Loyalty_Tier]="premium")`
- 10.no. of high tier customers=`COUNTROWS(FILTER(VALUES(Customer_Transactions[Customer_ID]),CALCULATE(COUNT(Customer_Transactions[Transaction_ID]))>8))`
- 11.no. of high value customers = `CALCULATE(COUNT(Customer_Demographics[Customer_ID]),Customer_Demographics[CLV]> 486.19)`
- 12.no. of low tier customers = `COUNTROWS(FILTER(VALUES(Customer_Transactions[Customer_ID]),CALCULATE(COUNT(Customer_Transactions[Transaction_ID]))<4))`
- 13.no. of low value customers = `CALCULATE(COUNT(Customer_Demographics[Customer_ID]),Customer_Demographics[CLV]< 486.19)`
- 14.no. of mid tier customers = `COUNTROWS(FILTER(VALUES(Customer_Transactions[Customer_ID]),CALCULATE(COUNT(Customer_Transactions[Transaction_ID]))<9))`
- 15.Non repeate customers = `COUNT(Customer_Demographics[Customer_ID]) - [repeate customers]`
- 16.point earn rate = `DIVIDE(SUM(Loyalty_Program[Points_Earned]), SUM(Customer_Transactions[Amount]),0)`
- 17.point earn rate = `DIVIDE(SUM(Loyalty_Program[Points_Earned]), SUM(Customer_Transactions[Amount]),0)`
- 18.points redemption rate = `DIVIDE(SUM(Loyalty_Program[Points_Redeemed]), SUM(Loyalty_Program[Points_Earned]),0)`
- 19.repeate customers = `COUNTROWS(FILTER(VALUES(Customer_Transactions[Customer_ID]),CALCULATE(COUNT(Customer_Transactions[Transaction_ID]))>2))`
- 20.retained customers = `CALCULATE(COUNT(Churn_Labelled_Customers[Churn_Flag]),Churn_Labelled_Customers[Churn_Flag]=0)`
- 21.retention rate = `DIVIDE(CALCULATE(COUNT(Churn_Labelled_Customers[Churn_Flag]),Churn_Labelled_Customers[Churn_Flag]=0),COUNT(Churn_Labelled_Customers[Churn_Flag]),0)`
22. total customers = `COUNT(Churn_Labelled_Customers[Customer_ID])`

Video link

(Link has be copied and pasted into browser because it is not opening by clicking)

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